

Physiological Signals Quality in 6DoF Virtual Reality: Preliminary Results

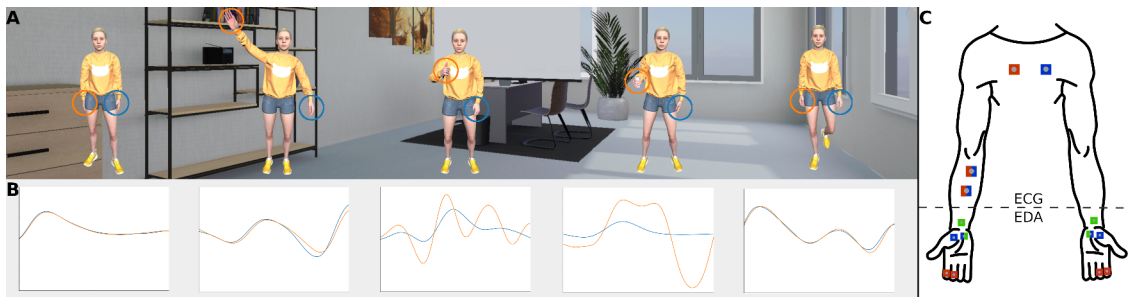
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Figure 1: A - 5 different gestures (Idle, Waving the hand, Thumbs up, Pick up and place an object, Walking); B - Corresponding EDA signals for both hands; C - Tested positions for EDA (Fingers, Palm and Wrist) and ECG (Chest Left and Chest Right)

ABSTRACT

Physiological data are increasingly used in Virtual Reality (VR) to analyze the user experience. However, when we move and interact in VR, the quality of physiological data is degraded. This study evaluates the impact of movements and sensor positions on signal quality. 3 positions for EDA and 2 for ECG have been compared on 36 subjects performing a series of different gestures in a within-subjects design. We observed that movements requiring the hand to be closed significantly degraded EDA signals, and that the palm position offered better robustness. For ECG, the position closest to the heart gave the best results.

Index Terms: Virtual Reality, Physiological Data, Signal Quality

1 INTRODUCTION

Today, many Virtual Reality (VR) studies seek to analyze and understand the user experience, the most common method being questionnaires. However, their use has limitations, and researchers are increasingly trying to combine such self-reported data with physiological signals and to find some links between these two types of data [1]. Several types of physiological data are considered in the literature, such as cardiovascular activity (ECG), electrodermal activity (EDA), neural activity, electromyography, eye tracking information, as well as respiration. ECG and EDA are the most commonly used in the literature from the VR field [2]. In a 6-Degrees-of-Freedom (6DoF) VR experiment, where the participant moves and interacts with the scene, the use of physiological data is however often compromised by a significant drop in the quality of the signals obtained. This is due to the electrical activity and sweating caused by movements, which directly modify ECG and EDA signals. These impairments require signal post-processing or sometimes the exclusion of part of the data. Wire movements can also create artifacts due to the triboelectric effect as well as pressing on

the sensors, which can happen when electrodes are placed on the user hand holding a controller.

The position of the sensors can influence the quality of the data. Several studies have proposed different alternatives, such as the forehead, the foot or the wrist for EDA [9]. Although some studies evaluate their noise resistance depending of movement [3] [7], none place themselves in a VR context in 6DoF with associated constraints such as the presence of controllers and headset and the realization of typical VR movements. A very recent position paper [5] indicates lack of knowledge in this context and raises the possibility of establishing a classification of signal artifacts according to the movements produced in virtual reality, but they only presented a research plan without experiment or results.

In this context, we propose to study the robustness of physiological signals in the frame of 6DoF virtual reality. More precisely, we focus here on evaluating the impact of different movements and sensor positions on data quality.

2 METHOD

2.1 Procedure and Task

We conducted a user study to investigate the effects of movement and position of physiological sensors on the quality of recorded signals. The positions compared here for EDA are the fingers, the wrist, and the palm (see Fig. 1). Finger position is the gold standard for collecting EDA signals, while wrist and palm are alternatives that minimize intrusiveness and offer sufficient power signals. For ECG, two positions are evaluated here with two electrodes on the right arm in common for the two positions, and a third electrode localized on the chest, either on the right or left side to keep the signal strong enough. The evaluated gestures are: Idle, Waving the hand, Thumbs up, Pick up and drop an object, Walking. These gestures cover typical movements in a 6DoF VR experience, but also different ranges of movement and cases where the hand is open or closed. The movements were performed with one moving hand and one stationary hand, alternating during each movement.

The experiment consists of four trials in a within-subjects design. The first three are used to evaluate the three EDA and two ECG positions while the last is reserved for comparing our EDA sensor with a Shimmer3 GRS+, which is among the most widely used in

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the literature, both positioned on the fingers of one hand. Subjects were equipped with one EDA sensor on each hand and a single ECG sensor. After a 5-minute waiting period for the skin-electrode contact to be effective, subjects were immersed alone in VR in a virtual office. A virtual agent stood in front of them and performed the series of five movements (See Fig. 1). The subjects' task was to imitate virtual agent's movements. The subjects were equipped with VR controllers as it reflects classical VR experiences.

We needed metrics to evaluate the signal quality. To quantify the amount of noise present in ECG signals, we use the template matching method commonly used to classify ECG signals [8]. Unlike ECG, the EDA signal does not have a known structure, which makes it difficult to detect noise in the signal. To overcome this, we considered the signal similarity (evaluated as Pearson correlation) between the moving hand and the stationary hand during each movement. This metric has already been used for signal evaluation [4], as the two signals are normally almost identical due to the lateralization of EDA. The comfort of use and the intrusiveness of each position were evaluated by a questionnaire after each trial.

2.2 Participants and Materials

36 volunteer subjects (23 male, 13 female) participated in the study (age 18-60, mean 33.3, SD 11.5). They were informed of the purpose of the experiment and the procedure. The VR headset used was an HTC VIVE Pro 2 with the Valve Index controllers. These were chosen because of their shape, which allows the electrodes to be placed on the palm, and because they offer excellent ergonomics. We used recently introduced EDA and ECG sensors [6] that have the advantage of being open-source, wireless, and using Bluetooth Low Energy¹. The EDA and ECG signals were sampled at a frequency of $F_s = 8$ Hz and $F_s = 512$ Hz respectively. The EDA signals were processed with a high-pass filter (Butterworth second order, cut-off 0.08 Hz) to remove the tonic part of the signal, which represents slow changes in the signal, leaving only the phasic component, which represents responses to emotional stimuli.

3 RESULTS

We run Friedman tests on our dependent variables of signal quality (See Sec. 2.1) according to sensor positions and gestures, and post-hoc tests. We observe no significant differences in EDA quality between the three positions ($p = 0.8455$). For gestures, we observe significant differences ($p < 0.001$). Idle and Walking gave the highest quality, then Waving slightly below, and finally Thumbs Up and Pickup/Place with the worst one. Finally, comparing the results of each position by movement (Fig. 2), we observe that the Finger position offers the best results for Walking, but the worst for the Pickup/Place, for which the Palm position provides the highest signal quality. For the ECG, the results about the position factor show that the Left Chest position offers significantly better signal quality than Chest Right ($P < 0.001$). Significant differences between movements were observed ($P < 0.001$). Idle movement gave the highest signal quality while Waving leads to the worst. Comparing the results of each position by movement (Fig. 2), we observe a consistently better quality in the Chest Left position.

4 CONCLUSION AND RECOMMENDATIONS

PickUp/Place and ThumbsUp movements affect the quality of EDA signals the most. Our results suggest the Palm position as a good alternative for the Fingers position, as it offers similar results for all movements, and even better for PickUp/Place. We recommend using the Chest Left position for ECG capture, to obtain good quality signals despite the presence of movement. Finally, the data from the user comfort questionnaires and the Shimmer sensor have not yet been analyzed, and will be treated as future work. This study thus did not evaluate the impact of sensors and participants' gender.

¹<https://github.com/sensors-inl/Nervous-Toolkit>

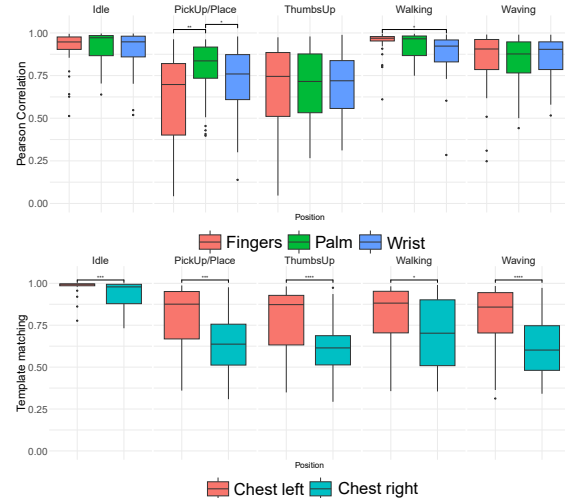


Figure 2: EDA and ECG signal quality between each position for each movement

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